The Effects of Blue Light on Anxiety-like and Depression-like Symptoms in Mouse Models

Emelia Delgado, Cass Delvira Vandevoorde, Jardyn Robinson, Dr. Stacey B. B. Dutton
Agnes Scott College
Department of Neuroscience

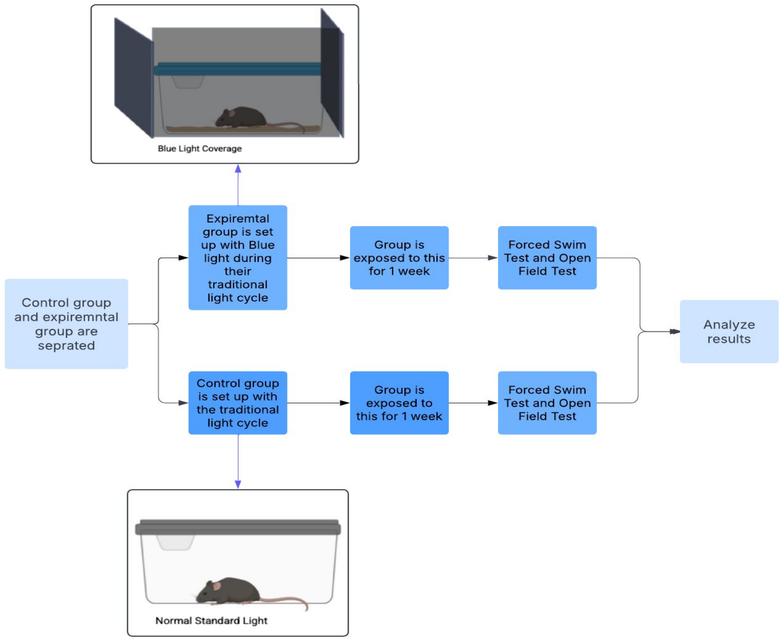
Abstract

Specific wavelengths of light are associated with different disruptions in circadian rhythm. Specifically, blue light has been shown to delay sleep onset. The purpose of this experiment was to observe the effects of anxiety-like and depression-like symptoms in mice. It was hypothesized that the blue light would increase anxiety-like and depression-like symptoms in experimental light. To complete this experiment, experimental mice were placed in blacked-out cages and exposed only to blue LED lights. The experimental mice were exposed to blue light 12 hours daily for one week before the forced swim and open field paradigms. Prediction: It is predicted that the experimental mice will spend less time in the perimeter in the open field paradigm and less time mobile in the forced swim paradigm, indicating that blue light increases anxiety-like and depression-like symptoms.

Introduction

Light plays a crucial role in regulating sleep-wake cycles and circadian rhythms. Research from the University of Oxford (2016) found that blue light exposure caused mice to fall asleep slower and also increased corticosterone, a stress hormone that can disrupt normal rhythms. In contrast, standard white light allowed for a stable sleep-wake cycle without hormonal disturbances. Based on this, we hypothesize that Jackson mice exposed to blue light will show higher stress levels, leading to increased time in the center for the open field test and less mobility in the forced swim test. Overall, we predict that blue light will elevate anxiety-like and depression-like responses.

Materials and Methods



Results

Figure 1:Daily 12 hr. Blue Light exposure has No Effect on Depression-like Symptoms in Rodents

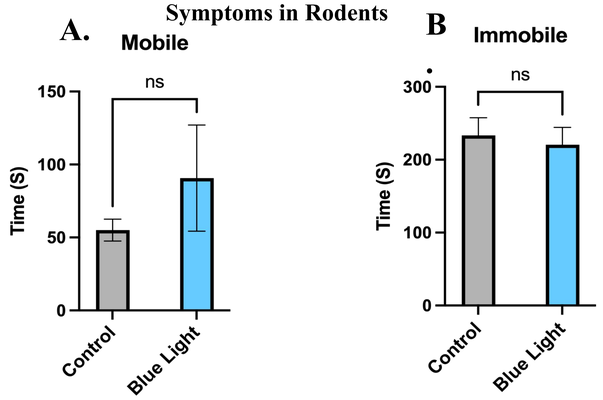


Figure 1: A. Illustrates time spent mobile for both groups. There was no significant difference; p = 0.4 B. Illustrates time spent immobile. There was no significant difference. These graphs illustrate that blue light has no effect on depression-like symptoms; p = 0.7 For each group, an N = 3 was used. Error bars represent SEM.

Results

Figure 2: Daily 12 hr. Blue Light exposure impacts Anxiety-like Symptoms in Rodents

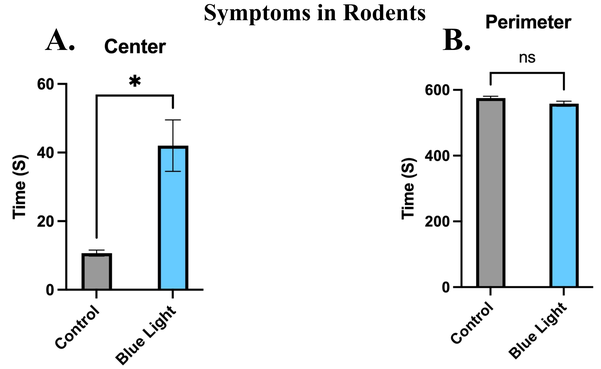


Figure 2:A. Illustrates time spent in the center for both groups. There is significant difference; p = 0.01. B. Illustrates time spent in outer sections. There was no significant difference; p = 0.14. These graphs illustrate that blue light has significant effect on decreasing anxiety-like symptoms. For each group, an N = 3 was used. Error bars represent SEM.

Discussions

- 12 hour blue light exposure does not impact depression-like symptoms
- 12 hour blue light exposure decreases anxiety-like symptoms
- Reduction in anxiety-like symptoms may be due to length of light exposure (Fonken et al., 2009)
- Lack of significance for time in perimeter may be due to variance or sample size (Cohen's d > 0.8)

Acknowledgement

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